System Architecture and Design

Architecture, data design and API reference (generated from the code)

Project title	IntelliMetrics AI: A Governed Semantic Layer with TARA, an AI Assistant for Trusted Business Metrics
Document	System Architecture and Design
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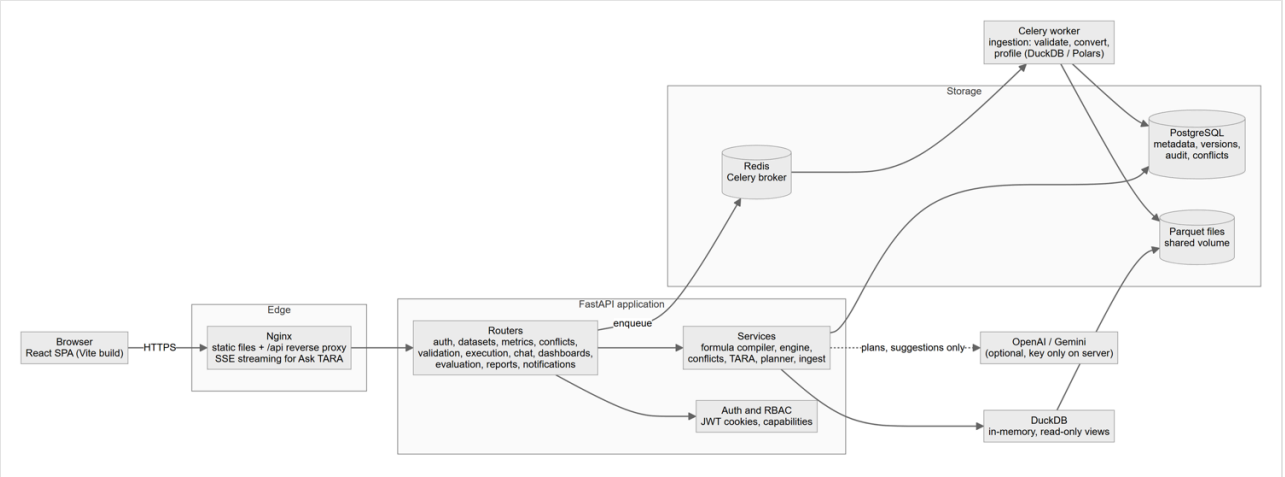
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1 Overview

IntelliMetrics AI is a three-layer web application with a separate background-job layer. This document describes the architecture with diagrams, the data design (ER diagram and data dictionary) and the API. The API reference, data dictionary, permission matrix and statistics in this document are **generated from the source code** by `docs/tools/gen_reference.py`; they are not hand-written, so they cannot drift from the implementation.

Fact (generated)	Value
API endpoints	96
Routers	16
Database tables	20
Role capabilities	19
Back-end source	46 Python files, 7,115 lines
Back-end tests	7 files, 1,072 lines
Front-end source	54 TypeScript files, 4,707 lines; 25 page components
End-to-end tests	2 files, 292 lines

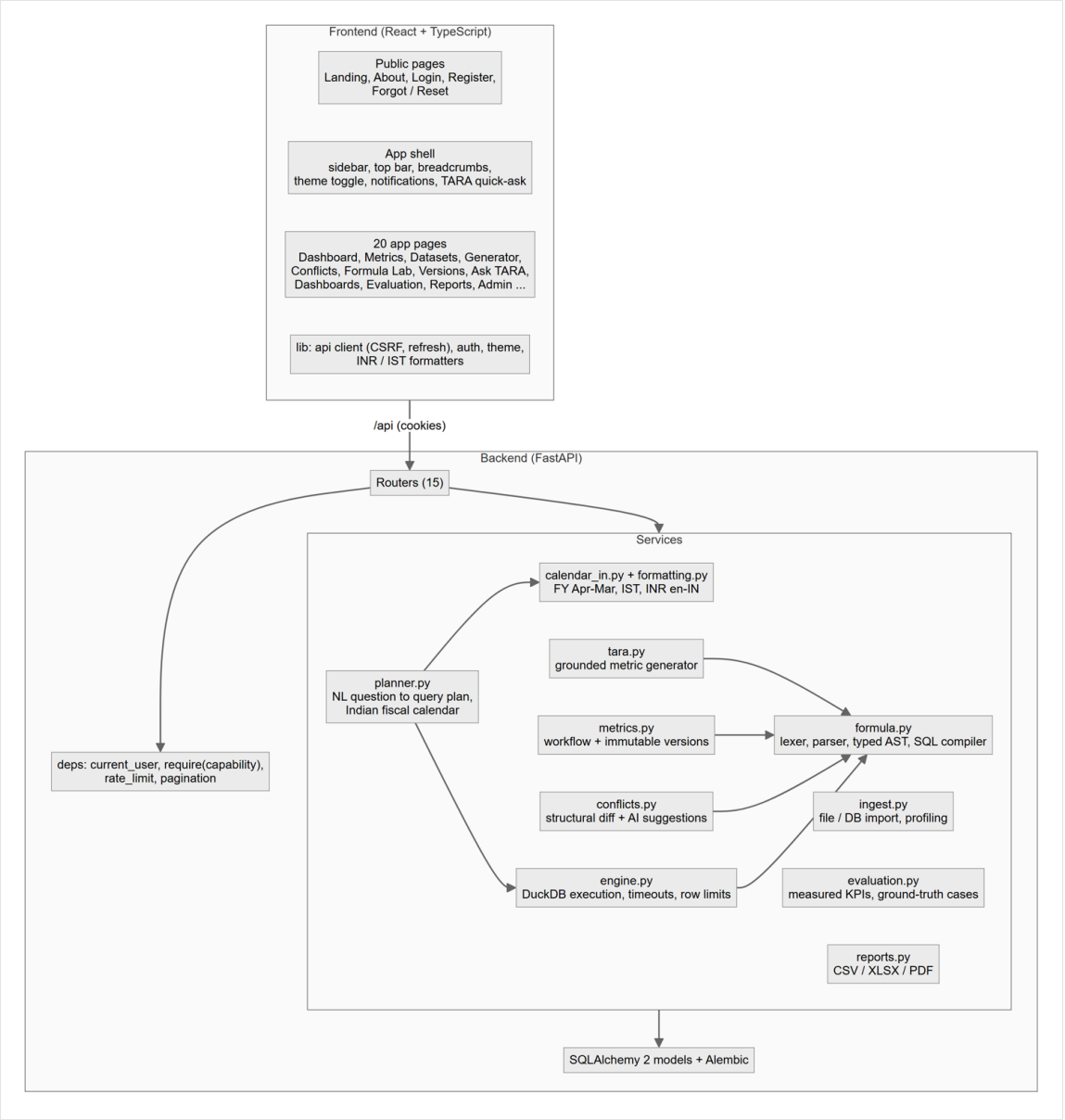
2 High-level architecture



High-level architecture

The browser talks only to Nginx (production) or the Vite dev proxy (development), so the SPA and API share one origin and the httpOnly cookies are first-party. FastAPI authenticates every request, authorises it by capability, filters every query by workspace, and delegates work to services. Long-running ingestion is placed on a Celery queue (Redis) and executed by a worker. Analytical queries are never sent to PostgreSQL: they run in DuckDB over Parquet files produced by ingestion.

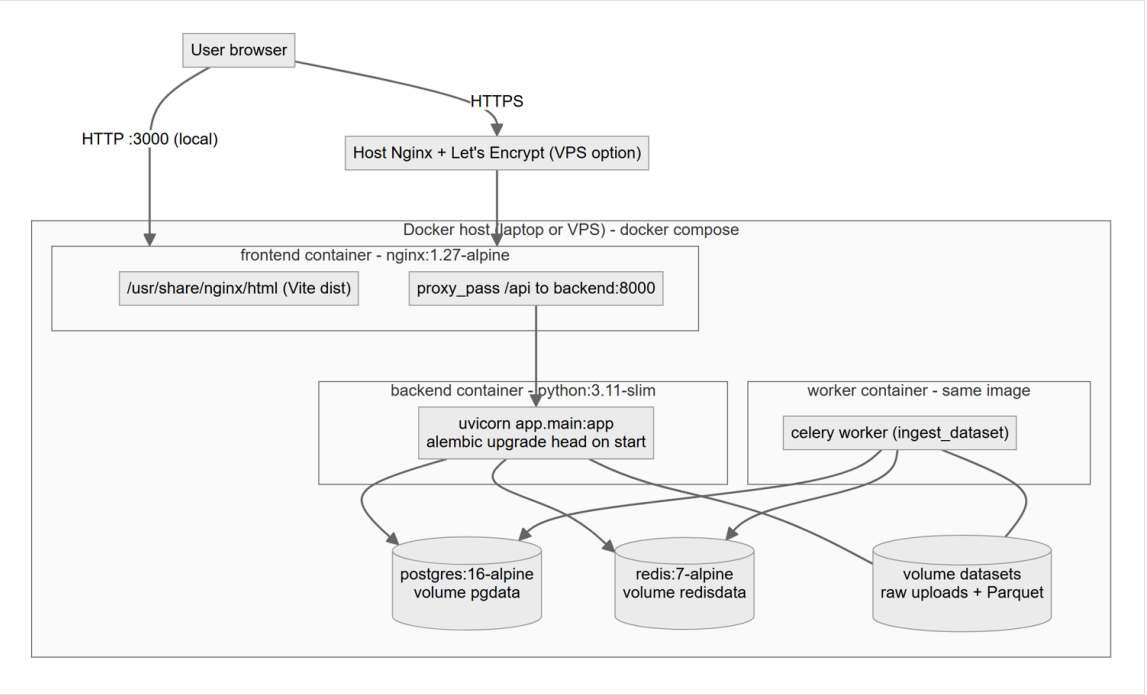
3 Component diagram



Components of the front end and back end

Responsibilities are layered so that safety-critical logic exists once: the **formula compiler** is used by the Formula Lab, metric validation, the TARA generator, the Ask TARA pipeline, dashboards and the evaluation module. The **planner** only produces a query plan; the **engine** executes it.

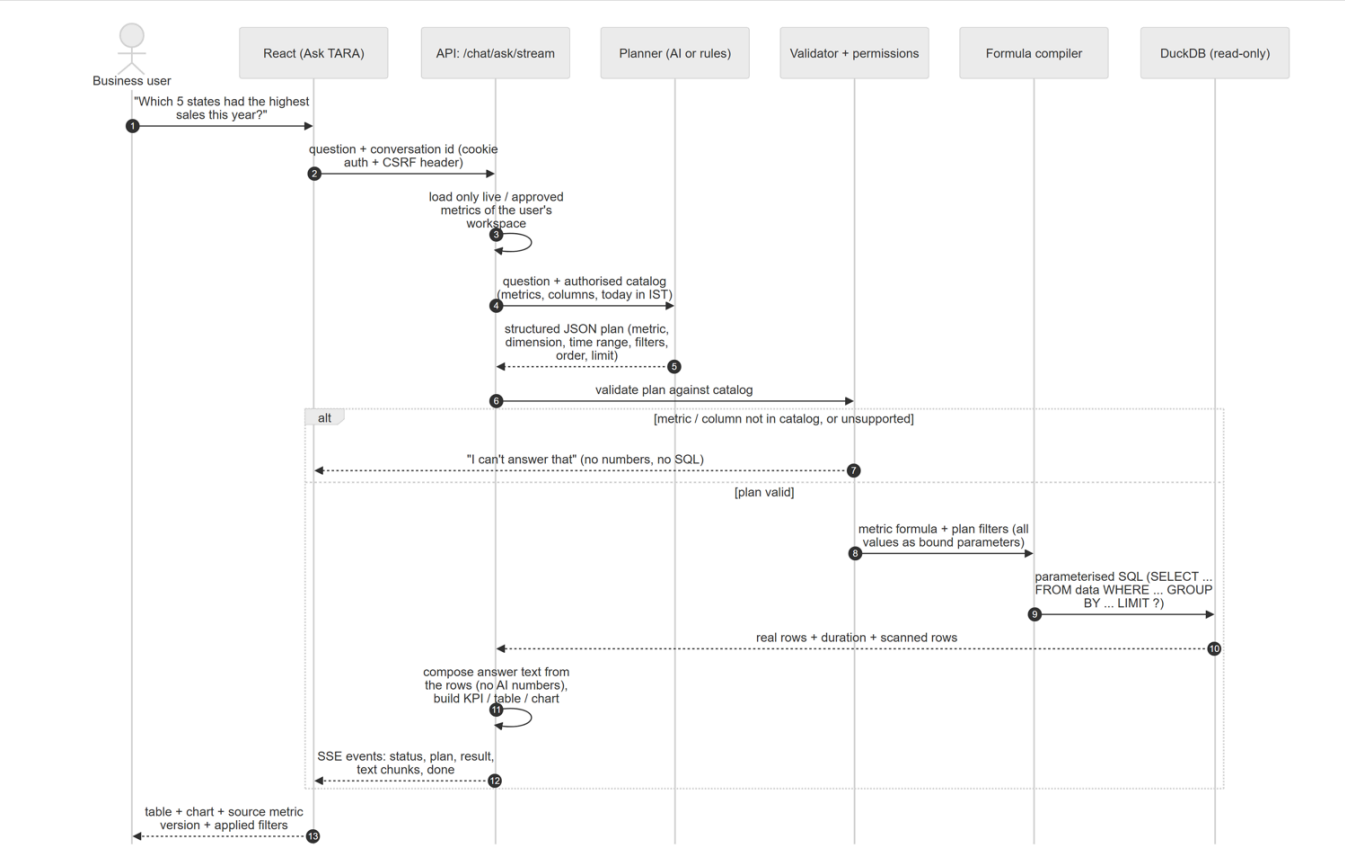
4 Deployment diagram



Docker Compose deployment (and optional VPS TLS terminator)

Persistent volumes: `pgdata` (PostgreSQL), `redisdata` (Redis append-only file) and `datasets` (raw uploads and Parquet), the last mounted in both the API and the worker so the worker's output is visible to the API. The backend container runs `alembic upgrade head` on every start and seeds the demonstration data when `AUTO_SEED=true` and the database is empty.

5 Request flow: Ask TARA

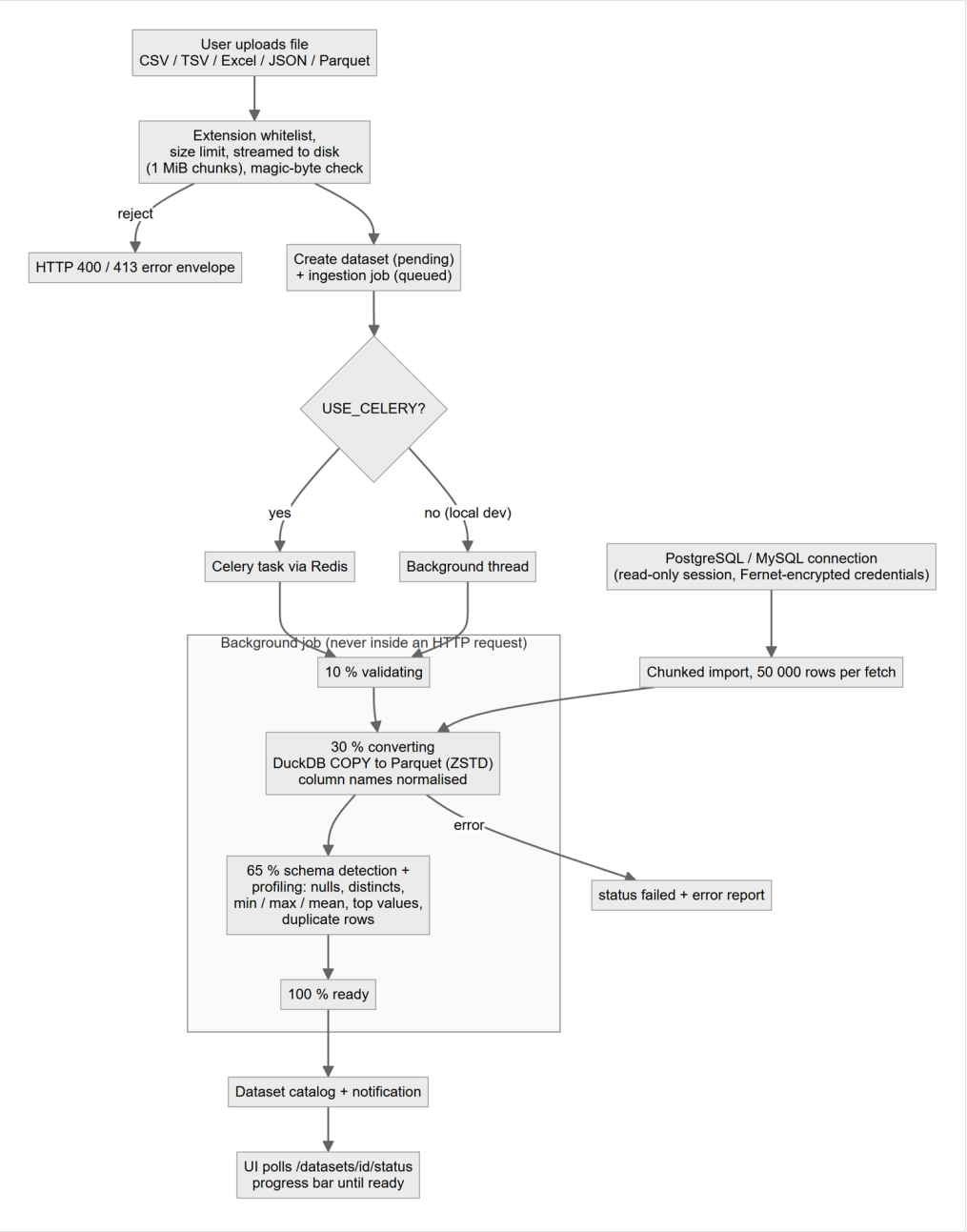


Sequence of an Ask TARA question

Key safety properties of this flow:

1. The planner receives only the **authorised catalog** (live or approved metrics of the user's own workspace on ready datasets) and today's date in IST.
2. Its output is a JSON *plan*, validated by Pydantic and then by `validate_plan`, which rejects unknown metrics or columns; a forged plan cannot reach the database.
3. The SQL is produced by the same compiler as in the Formula Lab; every value is a bound parameter.
4. The answer text is composed from the returned rows; the AI does not write the figures.

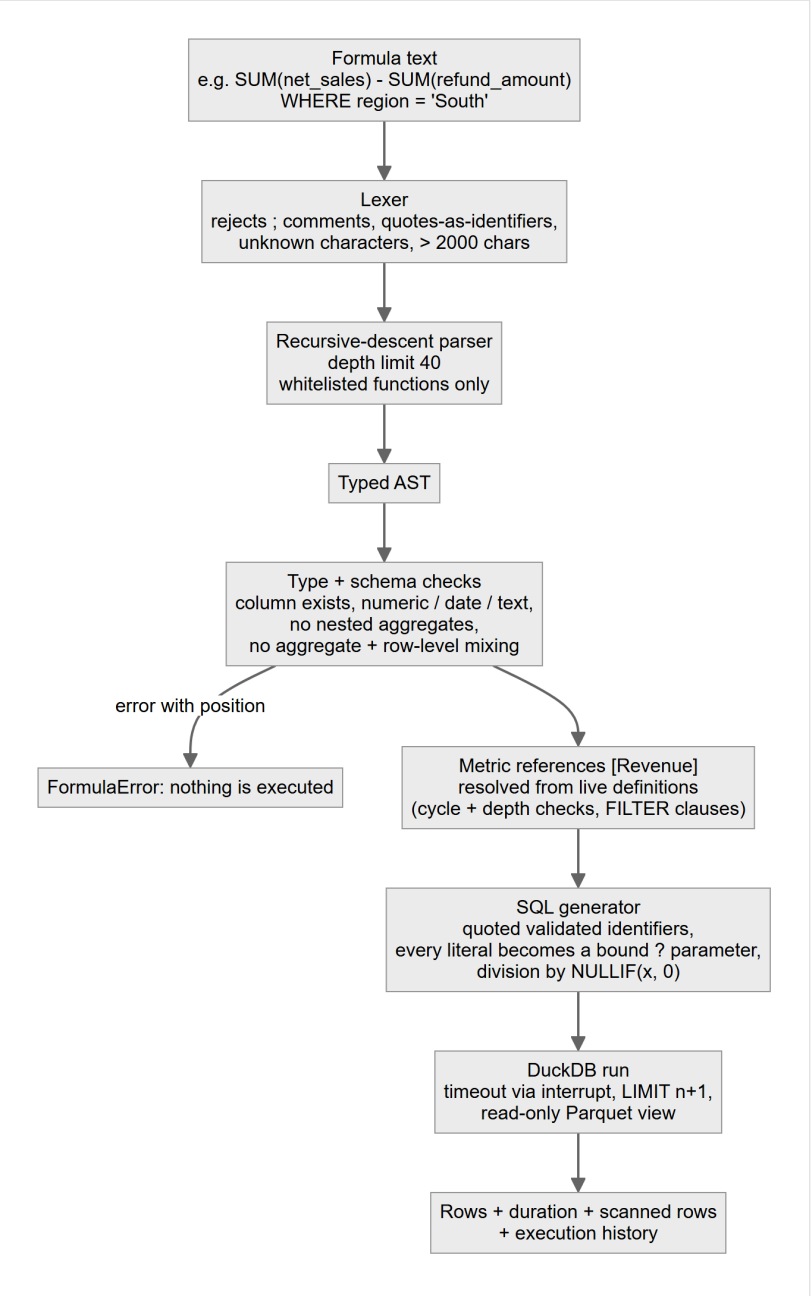
6 Dataset ingestion pipeline



Ingestion pipeline

Uploads are streamed to disk in 1 MiB chunks with an extension whitelist, a size limit and a magic-byte check. The job converts to ZSTD-compressed Parquet with DuckDB `COPY` (CSV rejects are recorded in an error report), normalises column names to snake_case, detects the schema and the default date column, and profiles every column in a single scan (null counts, approximate distinct counts, min, max, mean, standard deviation, top values) plus a duplicate-row count. External PostgreSQL/MySQL tables are streamed in 50,000-row chunks through a read-only session into a PyArrow Parquet writer.

7 Formula compile pipeline



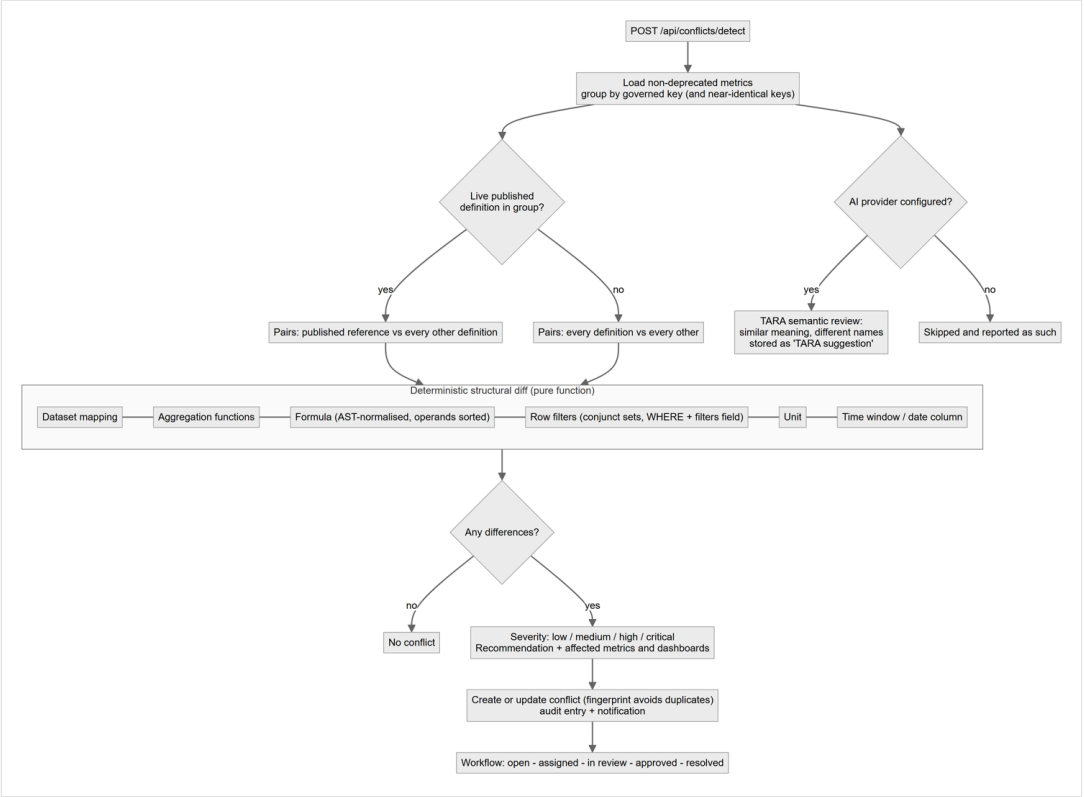
Formula lexer, parser, type checker and SQL generator

7.1 Grammar

```
formula   := expr [ WHERE bool ]
expr      := term (('+' | '-') term)*
term      := unary (('*' | '/' | '%') unary)*
unary     := '-' unary | primary
primary   := NUMBER | STRING | DATE STRING | TRUE | FALSE | NULL
           | '(' expr ')' | call | IDENT | '[' Metric Name ']'
call      := NAME '(' [ '*' | arg (',' arg)* ] ')'     -- whitelisted NAME only
bool      := and (OR and)* ;  and := not (AND not)* ;  not := [NOT] cmp
cmp       := expr (('='|'!='|'<>'|'<'|'<='|'>'|'>=') expr
                  | [NOT] IN '(' expr (',' expr)* ')' | [NOT] BETWEEN expr AND expr
                  | [NOT] LIKE STRING | IS [NOT] NULL) | '(' bool ')'
aggregates: SUM AVG MIN MAX COUNT COUNT_DISTINCT MEDIAN STDDEV SUMIF COUNTIF AVGIF
scalars:    ABS ROUND COALESCE IF NULLIF UPPER LOWER YEAR MONTH LENGTH
```

Limits: 2,000 characters, nesting depth 40, 200 items per `IN` list, 10,000 returned rows, 20-second timeout (all configurable). Semantic rules: columns must exist (a close-match suggestion is offered), `SUM/AVG/MEDIAN/STDDEV` require numeric input, text and date comparisons are type-checked, ISO date strings are coerced to dates, aggregates cannot be nested or mixed with row-level columns, a top-level formula must be a numeric aggregate expression, metric references must be live and defined on the same dataset.

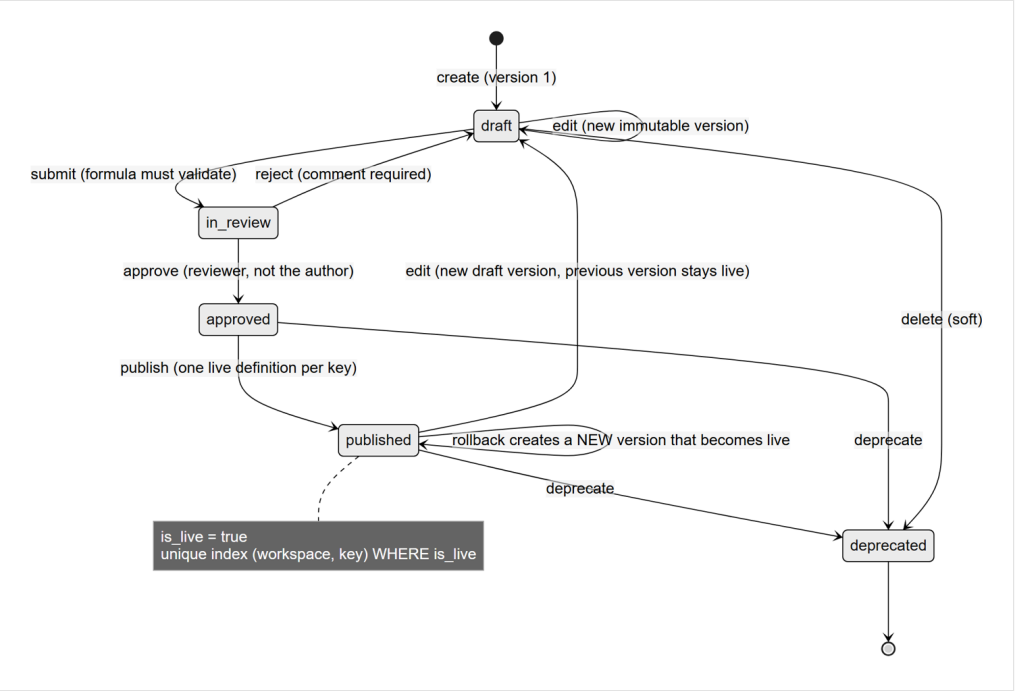
8 Conflict detection flow



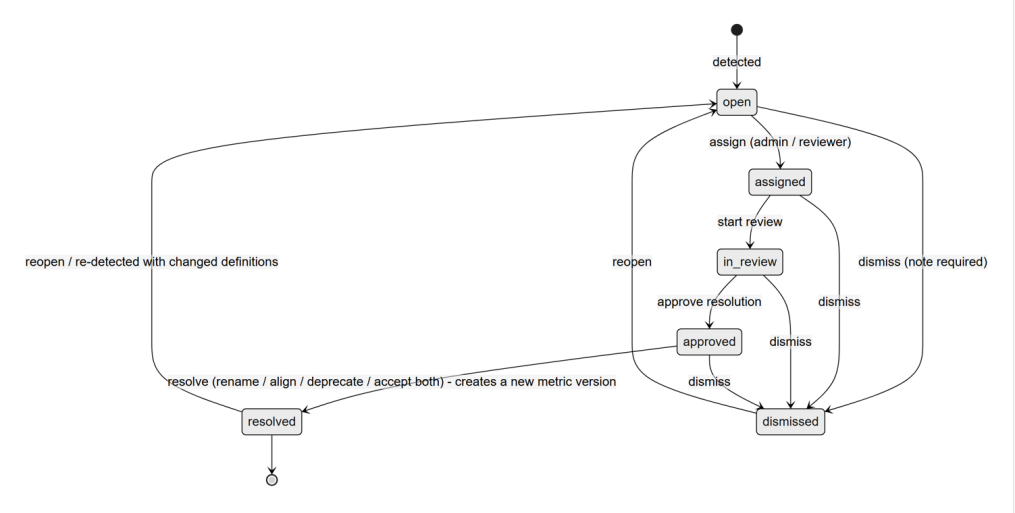
Conflict detection

Severity rules: the maximum severity of the differences, raised to *critical* when the reference is live and the variant is already in review/approved/published with a formula, aggregation or dataset difference, or when three or more difference types are present together with a high one.

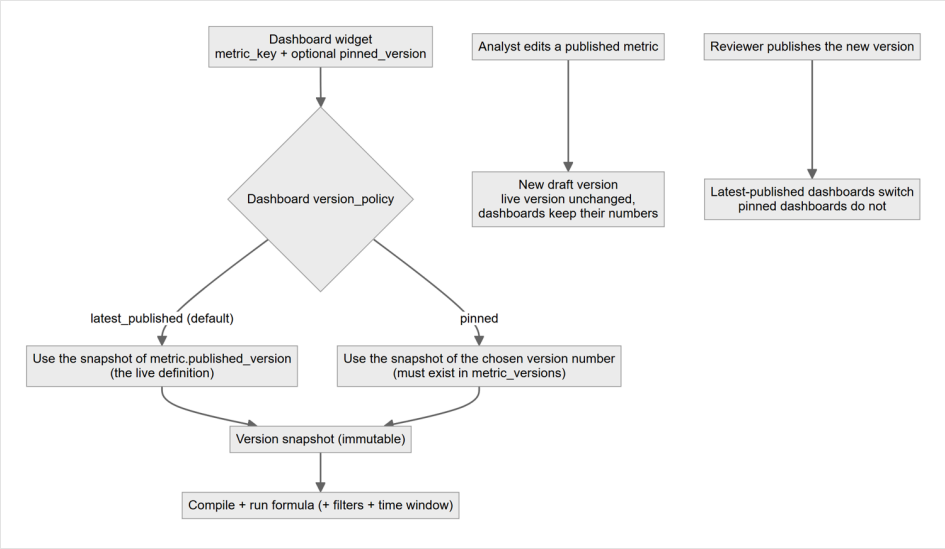
9 State machines



Metric lifecycle and version policy



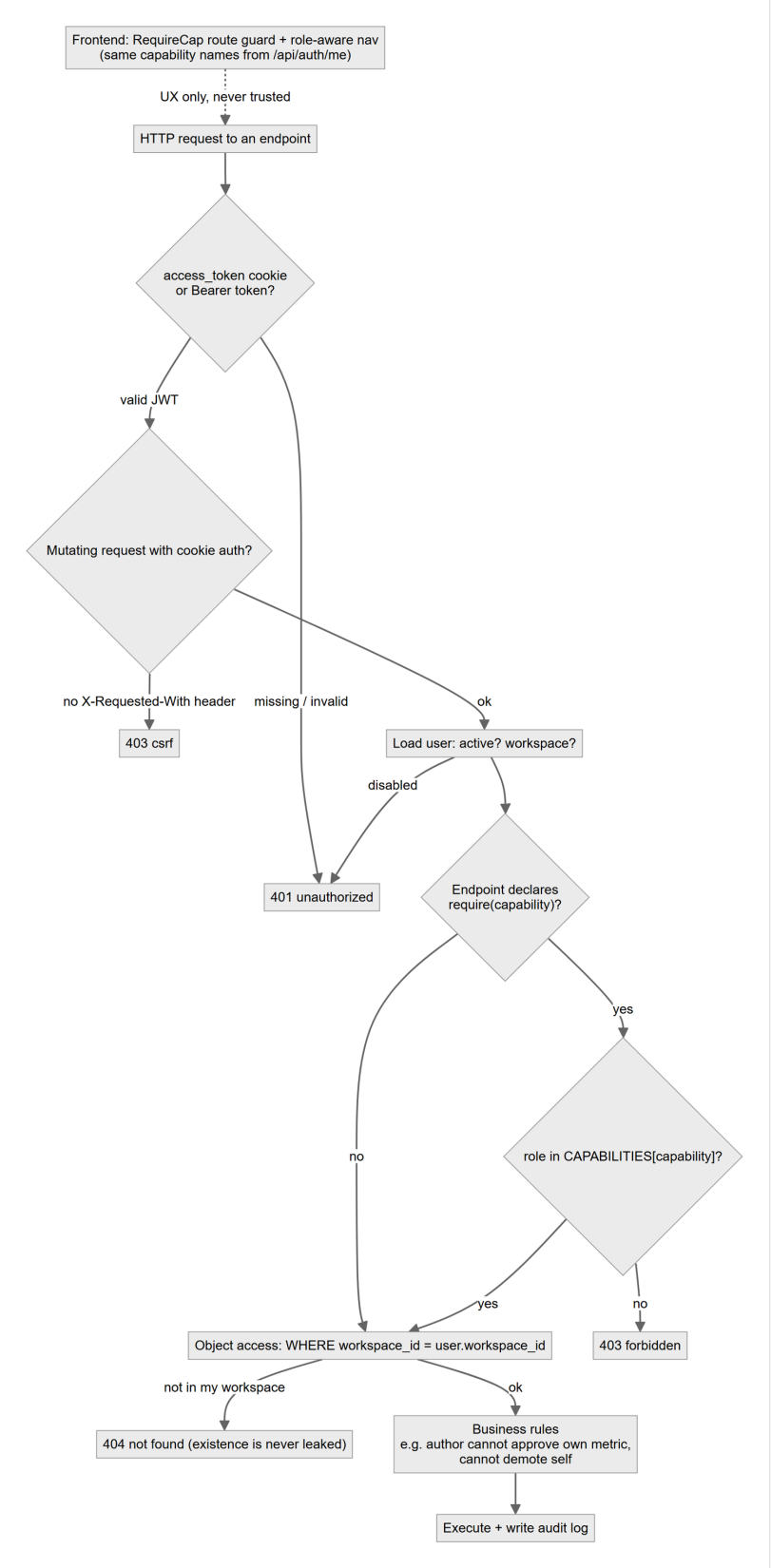
Conflict workflow



Dashboard version policy: latest published or pinned

10 Security architecture

10.1 Role-based access control



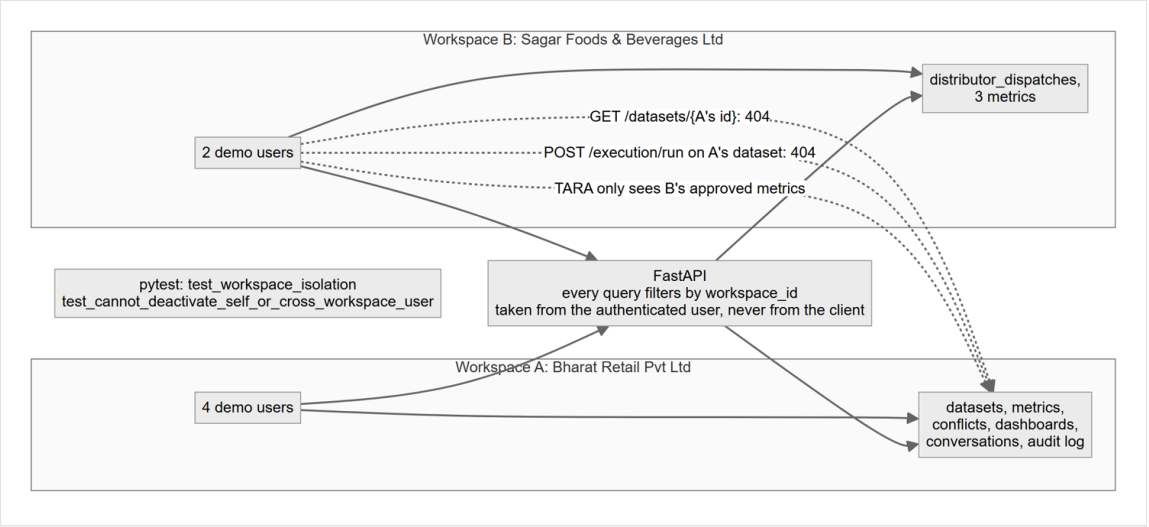
Permission flow for every request

Capability	Admin	Data Analyst	Business User	Reviewer
View metrics catalog, datasets, dashboards (`catalog.view`)	Yes	Yes	Yes	Yes
Upload datasets / manage connections (`dataset.write`)	Yes	Yes	-	-
Delete datasets and connections (`dataset.delete`)	Yes	-	-	-
Create and edit metrics (`metric.write`)	Yes	Yes	-	-
Approve / reject / publish / deprecate metrics (`metric.review`)	Yes	-	-	Yes
Roll back a metric (creates a new version) (`metric.rollback`)	Yes	-	-	Yes
Use TARA metric generator (`ai.generate`)	Yes	Yes	-	-
Ask TARA (natural-language BI) (`chat.ask`)	Yes	Yes	Yes	Yes
View conflicts (`conflict.view`)	Yes	Yes	Yes	Yes
Run conflict detection (`conflict.detect`)	Yes	Yes	-	Yes
Assign / review / resolve conflicts (`conflict.manage`)	Yes	-	-	Yes
Formula Lab: validate and execute (`formula.run`)	Yes	Yes	-	Yes
Create / edit dashboards (`dashboard.write`)	Yes	Yes	Yes	-
View performance evaluation (`evaluation.view`)	Yes	Yes	-	Yes
Run performance evaluation (`evaluation.run`)	Yes	Yes	-	-

Capability	Admin	Data Analyst	Business User	Reviewer
Export reports (`report.export`)	Yes	Yes	Yes	Yes
Manage users and roles (`user.manage`)	Yes	-	-	-
Manage workspace and settings (`workspace.manage`)	Yes	-	-	-
View audit log (`audit.view`)	Yes	-	-	-

Additional rules enforced in code: an author cannot approve their own metric (except an Admin); rejection requires a comment; users cannot demote or deactivate themselves; demo-account passwords cannot be changed; every sensitive action is audit-logged with user, entity, details, IP and request id.

10.2 Workspace isolation



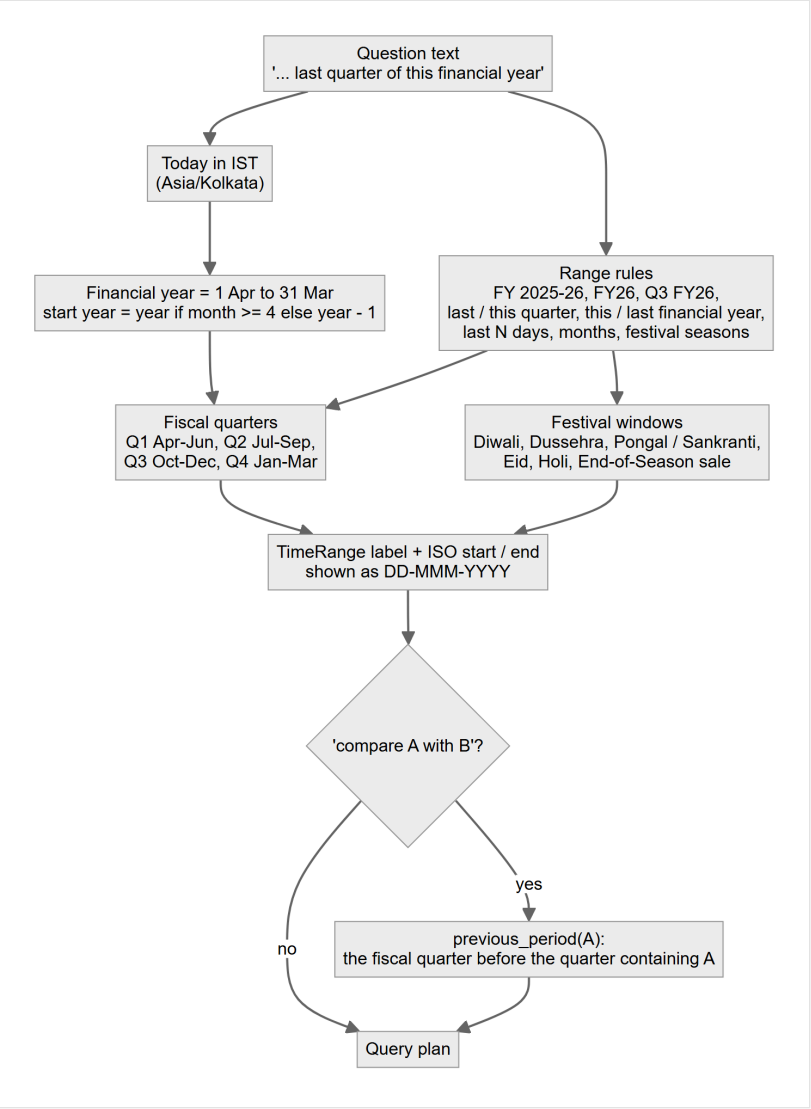
Workspace isolation

10.3 Other controls

Control	Implementation
Password storage	Argon2id hashes (argon2-cffi)
Sessions	JWT access token (15 min) and rotating refresh token (7 days, stored hashed) in httpOnly, SameSite cookies; `Secure` when `COOKIE_SECURE=true`
CSRF	State-changing cookie-authenticated requests must carry `X-Requested-With: IntelliMetrics`
Rate limiting	Sliding window per IP on authentication routes (per process)
Secrets	Environment variables only; connection passwords encrypted with Fernet; AI keys never reach the browser

Control	Implementation
Input validation	Pydantic v2 on every request; consistent error envelope with request id
SQL safety	No raw SQL accepted; restricted grammar; parameters; read-only DuckDB views; timeouts and row limits
Upload safety	Extension whitelist, size limit, magic-byte sniffing, binary rejection, stored outside the web root
Immutability	ORM guard plus database trigger on `metric_versions`

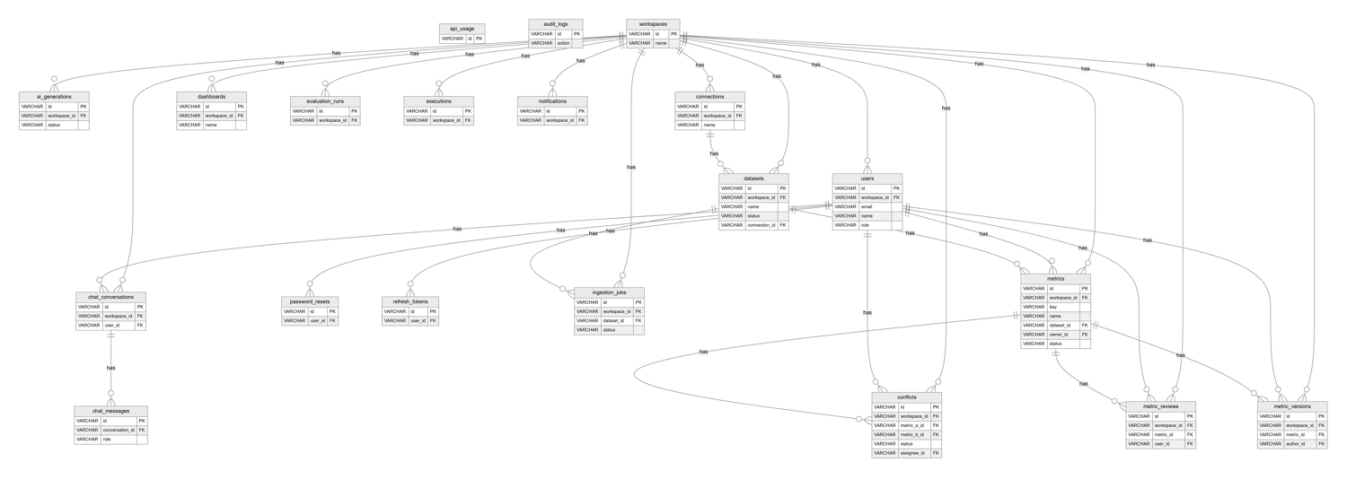
11 Indian calendar and planner



Fiscal calendar handling in the planner

12 Data design

12.1 Entity-relationship diagram



Entity-relationship diagram (keys and identifying columns; all columns are listed in the data dictionary)

The diagram is wide; for a legible full-size version open `docs/diagrams/er.svg` (zoomable vector) or `docs/diagrams/er.png`. Notes: every business table references `workspaces` (tenant boundary). `metrics` holds the working copy; `metric_versions` holds immutable snapshots; `metric_reviews` is the append-only approval log; `conflicts` references the reference metric and the variant; `dashboards.widgets` is a JSON layout whose entries reference metric keys and optional pinned version numbers; `ingestion_jobs` and `executions` are the telemetry that the evaluation module analyses.

12.2 Data dictionary

`api_usage`: AI provider call log used for usage statistics.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	no	indexed
`provider`	VARCHAR	no	
`kind`	VARCHAR	no	
`tokens_est`	INTEGER	no	
`latency_ms`	FLOAT	no	
`ok`	BOOLEAN	no	
`created_at`	DATETIME	no	

`audit_logs`: Audit trail of every sensitive action with user, entity, details, IP and request id.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	yes	indexed
`user_id`	VARCHAR	yes	
`user_email`	VARCHAR	yes	
`action`	VARCHAR	no	indexed
`entity_type`	VARCHAR	yes	indexed
`entity_id`	VARCHAR	yes	indexed
`details`	JSON	no	
`ip`	VARCHAR	yes	
`request_id`	VARCHAR	yes	
`created_at`	DATETIME	no	indexed

`workspaces` : Tenant boundary. Every other table carries a workspace_id; settings JSON holds AI provider, theme and retention.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`name`	VARCHAR	no	
`slug`	VARCHAR	no	unique
`is_demo`	BOOLEAN	no	
`settings`	JSON	no	
`created_at`	DATETIME	no	

`ai_generations` : TARA metric-generator suggestions with mode (ai/offline), validity, acceptance and latency.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	no	FK -> workspaces.id, indexed
`user_id`	VARCHAR	yes	
`prompt`	TEXT	no	
`dataset_id`	VARCHAR	yes	
`mode`	VARCHAR	no	
`provider`	VARCHAR	yes	
`latency_ms`	FLOAT	no	
`output`	JSON	no	
`valid`	BOOLEAN	no	
`status`	VARCHAR	no	
`metric_id`	VARCHAR	yes	

Column	Type	Null	Key / constraint
`accepted`	BOOLEAN	no	
`error`	TEXT	yes	
`created_at`	DATETIME	no	

`connections` : External PostgreSQL/MySQL connections; the password is Fernet-encrypted at rest.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	no	FK -> workspaces.id, indexed
`name`	VARCHAR	no	
`kind`	VARCHAR	no	
`host`	VARCHAR	no	
`port`	INTEGER	no	
`database`	VARCHAR	no	
`username`	VARCHAR	no	
`password_enc`	TEXT	no	
`options`	JSON	no	
`last_status`	VARCHAR	yes	
`last_error`	TEXT	yes	
`last_tested_at`	DATETIME	yes	
`created_by`	VARCHAR	yes	
`created_at`	DATETIME	no	

`dashboards` : Saved dashboards: widget layout JSON, filters and the version policy (latest_published or pinned).

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	no	FK -> workspaces.id, indexed
`name`	VARCHAR	no	
`description`	TEXT	yes	
`widgets`	JSON	no	
`filters`	JSON	no	
`version_policy`	VARCHAR	no	
`template`	VARCHAR	yes	
`created_by`	VARCHAR	yes	
`created_at`	DATETIME	no	
`updated_at`	DATETIME	no	

`evaluation_runs` : Stored results of each performance evaluation run.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	no	FK -> workspaces.id, indexed
`user_id`	VARCHAR	yes	
`results`	JSON	no	
`duration_ms`	FLOAT	no	
`created_at`	DATETIME	no	

`executions` : History of validations and executions (Formula Lab, Ask TARA, dashboards) with duration, scanned rows and error.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	no	FK -> workspaces.id, indexed
`user_id`	VARCHAR	yes	
`dataset_id`	VARCHAR	yes	
`kind`	VARCHAR	no	
`expression`	TEXT	yes	
`ok`	BOOLEAN	no	
`duration_ms`	FLOAT	no	
`scanned_rows`	BIGINT	no	
`row_count`	BIGINT	no	
`error`	TEXT	yes	
`sql_text`	TEXT	yes	
`created_at`	DATETIME	no	indexed

`notifications` : In-app notifications per workspace or user with per-user read state.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	no	FK -> workspaces.id, indexed
`user_id`	VARCHAR	yes	indexed
`title`	VARCHAR	no	
`body`	TEXT	yes	
`kind`	VARCHAR	no	
`link`	VARCHAR	yes	
`read_by`	JSON	no	

Column	Type	Null	Key / constraint
`created_at`	DATETIME	no	indexed

`users` : Accounts with an Argon2 password hash, a role (admin, data_analyst, business_user, reviewer) and a workspace.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	no	FK -> workspaces.id, indexed
`email`	VARCHAR	no	unique
`name`	VARCHAR	no	
`password_hash`	VARCHAR	no	
`role`	VARCHAR	no	
`is_active`	BOOLEAN	no	
`is_demo`	BOOLEAN	no	
`avatar_url`	VARCHAR	yes	
`title`	VARCHAR	yes	
`created_at`	DATETIME	no	
`last_login`	DATETIME	yes	

`chat_conversations` : Ask TARA conversations per user.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	no	FK -> workspaces.id, indexed
`user_id`	VARCHAR	no	FK -> users.id, indexed
`title`	VARCHAR	no	
`created_at`	DATETIME	no	
`updated_at`	DATETIME	no	

`datasets` : Dataset catalog: format, status/progress, detected schema, column profile and the Parquet path produced by ingestion.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	no	FK -> workspaces.id, indexed
`name`	VARCHAR	no	
`description`	TEXT	yes	
`source_type`	VARCHAR	no	
`file_format`	VARCHAR	yes	

Column	Type	Null	Key / constraint
`original_filename`	VARCHAR	yes	
`raw_path`	VARCHAR	yes	
`parquet_path`	VARCHAR	yes	
`size_bytes`	BIGINT	no	
`row_count`	BIGINT	no	
`column_count`	INTEGER	no	
`status`	VARCHAR	no	
`progress`	INTEGER	no	
`stage`	VARCHAR	yes	
`error`	TEXT	yes	
`schema_json`	JSON	no	
`profile`	JSON	no	
`date_column`	VARCHAR	yes	
`connection_id`	VARCHAR	yes	FK -> connections.id
`source_table`	VARCHAR	yes	
`is_demo`	BOOLEAN	no	
`created_by`	VARCHAR	yes	
`created_at`	DATETIME	no	
`updated_at`	DATETIME	no	

`password_resets` : Single-use, hashed, one-hour password reset tokens.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`user_id`	VARCHAR	no	FK -> users.id, indexed
`token_hash`	VARCHAR	no	unique
`expires_at`	DATETIME	no	
`used`	BOOLEAN	no	
`created_at`	DATETIME	no	

`refresh_tokens` : Hashed, rotating refresh tokens (httpOnly cookie) with expiry and revocation flag.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`user_id`	VARCHAR	no	FK -> users.id, indexed
`token_hash`	VARCHAR	no	unique
`expires_at`	DATETIME	no	

Column	Type	Null	Key / constraint
`revoked`	BOOLEAN	no	
`created_at`	DATETIME	no	

`chat_messages` : Messages with the structured answer payload (table, chart, sources, applied filters).

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`conversation_id`	VARCHAR	no	FK -> chat_conversations.id, indexed
`role`	VARCHAR	no	
`content`	TEXT	no	
`payload`	JSON	no	
`created_at`	DATETIME	no	

`ingestion_jobs` : One row per background ingestion run: stage, progress, duration, runner and error report.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	no	FK -> workspaces.id, indexed
`dataset_id`	VARCHAR	no	FK -> datasets.id, indexed
`status`	VARCHAR	no	
`stage`	VARCHAR	yes	
`progress`	INTEGER	no	
`rows`	INTEGER	no	
`bytes`	BIGINT	no	
`duration_ms`	INTEGER	yes	
`error_report`	JSON	no	
`runner`	VARCHAR	yes	
`created_at`	DATETIME	no	
`finished_at`	DATETIME	yes	

`metrics` : Governed metric definitions (working copy). A partial unique index allows only one live definition per workspace and key.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	no	FK -> workspaces.id, indexed
`key`	VARCHAR	no	indexed
`name`	VARCHAR	no	

Column	Type	Null	Key / constraint
description	TEXT	yes	
business_definition	TEXT	yes	
formula	TEXT	no	
filters	TEXT	yes	
aggregation	VARCHAR	yes	
unit	VARCHAR	yes	
time_window	VARCHAR	no	
date_column	VARCHAR	yes	
dataset_id	VARCHAR	yes	FK -> datasets.id
department	VARCHAR	yes	
category	VARCHAR	yes	
owner_id	VARCHAR	yes	FK -> users.id
status	VARCHAR	no	
version	INTEGER	no	
published_version	INTEGER	yes	
is_live	BOOLEAN	no	
depends_on	JSON	no	
source	VARCHAR	no	
created_at	DATETIME	no	
updated_at	DATETIME	no	

Unique index `uq_metric_live_key` on (workspace_id, key) WHERE is_live IS TRUE (partial)

`conflicts` : Detected definition conflicts with differences, severity, recommendation, affected objects and workflow status.

Column	Type	Null	Key / constraint
id	VARCHAR	no	PK
workspace_id	VARCHAR	no	FK -> workspaces.id, indexed
metric_key	VARCHAR	no	indexed
metric_a_id	VARCHAR	no	FK -> metrics.id
metric_b_id	VARCHAR	no	FK -> metrics.id
conflict_type	VARCHAR	no	
severity	VARCHAR	no	
differences	JSON	no	
explanation	TEXT	no	
recommendation	TEXT	no	

Column	Type	Null	Key / constraint
`recommended_action`	VARCHAR	yes	
`affected`	JSON	no	
`fingerprint`	VARCHAR	no	
`status`	VARCHAR	no	
`assignee_id`	VARCHAR	yes	FK -> users.id
`resolution_action`	VARCHAR	yes	
`resolution_note`	TEXT	yes	
`detected_at`	DATETIME	no	
`resolved_at`	DATETIME	yes	

`metric_reviews` : Append-only approval workflow events: submit, approve, reject, publish, deprecate.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	no	FK -> workspaces.id, indexed
`metric_id`	VARCHAR	no	FK -> metrics.id, indexed
`version`	INTEGER	no	
`user_id`	VARCHAR	yes	FK -> users.id
`decision`	VARCHAR	no	
`comment`	TEXT	yes	
`created_at`	DATETIME	no	

`metric_versions` : Immutable version snapshots. UPDATE/DELETE are blocked by an ORM guard and a database trigger.

Column	Type	Null	Key / constraint
`id`	VARCHAR	no	PK
`workspace_id`	VARCHAR	no	FK -> workspaces.id, indexed
`metric_id`	VARCHAR	no	FK -> metrics.id, indexed
`number`	INTEGER	no	
`snapshot`	JSON	no	
`change_summary`	VARCHAR	no	
`reason`	TEXT	yes	
`author_id`	VARCHAR	yes	FK -> users.id
`rollback_of`	INTEGER	yes	
`created_at`	DATETIME	no	

13 API reference summary

All endpoints are under `/api` (health at `/health`); interactive OpenAPI documentation is served at `/docs` and the schema at `/openapi.json`. Errors use one envelope: `{"error": {"code", "message", "details", "request_id"}}`. The *Roles allowed* column is derived from the `require(capability)` dependency on each route and the permission matrix.

Authentication (`auth` router)

Method	Path	Purpose	Capability	Roles allowed
POST	`/api/auth/change-password`	change password	-	any signed-in user
POST	`/api/auth/forgot-password`	forgot password	-	public (no login)
POST	`/api/auth/login`	login	-	public (no login)
POST	`/api/auth/logout`	logout	-	public (no login)
GET	`/api/auth/me`	me	-	any signed-in user
PATCH	`/api/auth/me`	update me	-	any signed-in user
POST	`/api/auth/refresh`	refresh	-	public (no login)
POST	`/api/auth/register`	register	-	public (no login)
POST	`/api/auth/reset-password`	reset password	-	public (no login)

Workspaces, settings, audit log (`workspaces` router)

Method	Path	Purpose	Capability	Roles allowed
GET	`/api/admin/api-usage`	api usage	`workspace.manage`	Admin
GET	`/api/admin/audit-logs`	audit logs	`audit.view`	Admin
GET	`/api/permissions/matrix`	permissions matrix	-	any signed-in user
GET	`/api/workspaces/current`	get workspace	-	any signed-in user
PATCH	`/api/workspaces/current`	rename workspace	`workspace.manage`	Admin
GET	`/api/workspaces/current/settings`	get settings	-	any signed-in user
PUT	`/api/workspaces/current/settings`	put settings	`workspace.manage`	Admin

Users and roles (`users` router)

Method	Path	Purpose	Capability	Roles allowed
GET	`/api/users`	list users	-	any signed-in user
POST	`/api/users`	invite user	`user.manage`	Admin
DELETE	`/api/users/{user_id}`	remove user	`user.manage`	Admin

Method	Path	Purpose	Capability	Roles allowed
PATCH	`/api/users/{user_id}`	patch user	`user.manage`	Admin

Datasets, uploads and connections (`datasets` router)

Method	Path	Purpose	Capability	Roles allowed
GET	`/api/connections`	list connections	-	any signed-in user
POST	`/api/connections`	create connection	`dataset.write`	Admin, Analyst
DELETE	`/api/connections/{cid}`	delete conn	`dataset.delete`	Admin
POST	`/api/connections/{cid}/import`	conn import	`dataset.write`	Admin, Analyst
GET	`/api/connections/{cid}/tables`	conn tables	`dataset.write`	Admin, Analyst
POST	`/api/connections/{cid}/test`	test conn	`dataset.write`	Admin, Analyst
GET	`/api/datasets`	list datasets	-	any signed-in user
POST	`/api/datasets/upload`	upload dataset	`dataset.write`	Admin, Analyst
DELETE	`/api/datasets/{dataset_id}`	delete dataset	`dataset.delete`	Admin
GET	`/api/datasets/{dataset_id}`	get dataset	-	any signed-in user
GET	`/api/datasets/{dataset_id}/jobs`	dataset jobs	-	any signed-in user
GET	`/api/datasets/{dataset_id}/preview`	preview	-	any signed-in user
POST	`/api/datasets/{dataset_id}/reprocess`	reprocess	`dataset.write`	Admin, Analyst
GET	`/api/datasets/{dataset_id}/status`	dataset status	-	any signed-in user

Metrics, workflow and versions (`metrics` router)

Method	Path	Purpose	Capability	Roles allowed
GET	`/api/metrics`	list metrics	-	any signed-in user
POST	`/api/metrics`	create	`metric.write`	Admin, Analyst
GET	`/api/metrics/facets`	facets	-	any signed-in user
GET	`/api/metrics/graph`	graph	-	any signed-in user
DELETE	`/api/metrics/{mid}`	delete	`metric.write`	Admin, Analyst
GET	`/api/metrics/{mid}`	detail	-	any signed-in user
PATCH	`/api/metrics/{mid}`	update	`metric.write`	Admin, Analyst
POST	`/api/metrics/{mid}/approve`	metric approve	`metric.review`	Admin, Reviewer
GET	`/api/metrics/{mid}/audit`	metric audit	-	any signed-in user
POST	`/api/metrics/{mid}/deprecate`	metric deprecate	`metric.review`	Admin, Reviewer
GET	`/api/metrics/{mid}/diff`	diff	-	any signed-in user
POST	`/api/metrics/{mid}/publish`	metric publish	`metric.review`	Admin, Reviewer
POST	`/api/metrics/{mid}/reject`	metric reject	`metric.review`	Admin, Reviewer
POST	`/api/metrics/{mid}/rollback`	rollback	`metric.rollback`	Admin, Reviewer

Method	Path	Purpose	Capability	Roles allowed
POST	`/api/metrics/{mid}/submit`	metric submit	`metric.write`	Admin, Analyst
GET	`/api/metrics/{mid}/versions`	versions	-	any signed-in user
GET	`/api/versions/recent`	recent versions	-	any signed-in user

TARA metric generator (`ai` router)

Method	Path	Purpose	Capability	Roles allowed
POST	`/api/ai/generate-metric`	generate	`ai.generate`	Admin, Analyst
GET	`/api/ai/generations`	generations	`ai.generate`	Admin, Analyst
POST	`/api/ai/generations/{gid}/discard`	discard	`ai.generate`	Admin, Analyst
POST	`/api/ai/generations/{gid}/save`	save	`metric.write`	Admin, Analyst
GET	`/api/ai/status`	status	-	any signed-in user

Conflict detection (`conflicts` router)

Method	Path	Purpose	Capability	Roles allowed
GET	`/api/conflicts`	list conflicts	`conflict.view`	Admin, Analyst, Business, Reviewer
POST	`/api/conflicts/detect`	detect	`conflict.detect`	Admin, Analyst, Reviewer
GET	`/api/conflicts/{cid}`	detail	`conflict.view`	Admin, Analyst, Business, Reviewer
POST	`/api/conflicts/{cid}/approve`	approve	`conflict.manage`	Admin, Reviewer
POST	`/api/conflicts/{cid}/assign`	assign	`conflict.manage`	Admin, Reviewer
POST	`/api/conflicts/{cid}/dismiss`	dismiss	`conflict.manage`	Admin, Reviewer
GET	`/api/conflicts/{cid}/events`	events	`conflict.view`	Admin, Analyst, Business, Reviewer
POST	`/api/conflicts/{cid}/reopen`	reopen	`conflict.manage`	Admin, Reviewer
POST	`/api/conflicts/{cid}/resolve`	resolve	`conflict.manage`	Admin, Reviewer
POST	`/api/conflicts/{cid}/review`	start review	`conflict.manage`	Admin, Reviewer

Formula validation (`validation` router)

Method	Path	Purpose	Capability	Roles allowed
POST	`/api/validation/formula`	validate formula	`formula.run`	Admin, Analyst, Reviewer

Formula execution (`execution` router)

Method	Path	Purpose	Capability	Roles allowed
GET	`/api/execution/history`	history	`formula.run`	Admin, Analyst, Reviewer

Method	Path	Purpose	Capability	Roles allowed
POST	`/api/execution/run`	run	`formula.run`	Admin, Analyst, Reviewer

Ask TARA (`chat` router)

Method	Path	Purpose	Capability	Roles allowed
POST	`/api/chat/ask`	ask	`chat.ask`	Admin, Analyst, Business, Reviewer
POST	`/api/chat/ask/stream`	ask stream	`chat.ask`	Admin, Analyst, Business, Reviewer
GET	`/api/chat/conversations`	conversations	`chat.ask`	Admin, Analyst, Business, Reviewer
DELETE	`/api/chat/conversations/{cid}`	delete conversation	`chat.ask`	Admin, Analyst, Business, Reviewer
GET	`/api/chat/conversations/{cid}`	conversation	`chat.ask`	Admin, Analyst, Business, Reviewer
GET	`/api/chat/suggestions`	get suggestions	`chat.ask`	Admin, Analyst, Business, Reviewer

Dashboards (`dashboards` router)

Method	Path	Purpose	Capability	Roles allowed
GET	`/api/dashboards`	list dashboards	-	any signed-in user
POST	`/api/dashboards`	create	`dashboard.write`	Admin, Analyst, Business
POST	`/api/dashboards/from-template`	from template	`dashboard.write`	Admin, Analyst, Business
GET	`/api/dashboards/templates`	templates	-	any signed-in user
POST	`/api/dashboards/widget-data`	preview widget	-	any signed-in user
DELETE	`/api/dashboards/{did}`	delete	`dashboard.write`	Admin, Analyst, Business
GET	`/api/dashboards/{did}`	get	-	any signed-in user
PATCH	`/api/dashboards/{did}`	update	`dashboard.write`	Admin, Analyst, Business
POST	`/api/dashboards/{did}/data`	data	-	any signed-in user

Evaluation (`evaluation` router)

Method	Path	Purpose	Capability	Roles allowed
GET	`/api/evaluation/cases`	cases	`evaluation.view`	Admin, Analyst, Reviewer
GET	`/api/evaluation/latest`	latest	`evaluation.view`	Admin, Analyst, Reviewer

Method	Path	Purpose	Capability	Roles allowed
POST	`/api/evaluation/run`	run	`evaluation.run`	Admin, Analyst
GET	`/api/evaluation/runs`	runs	`evaluation.view`	Admin, Analyst, Reviewer
GET	`/api/evaluation/runs/{rid}`	get run	`evaluation.view`	Admin, Analyst, Reviewer

Reports and exports (`reports` router)

Method	Path	Purpose	Capability	Roles allowed
GET	`/api/reports/{kind}`	export	`report.export`	Admin, Analyst, Business, Reviewer

Notifications and activity (`notifications` router)

Method	Path	Purpose	Capability	Roles allowed
GET	`/api/activity`	activity	-	any signed-in user
GET	`/api/notifications`	list notifications	-	any signed-in user
POST	`/api/notifications/read-all`	mark all	-	any signed-in user
POST	`/api/notifications/{nid}/read`	mark read	-	any signed-in user

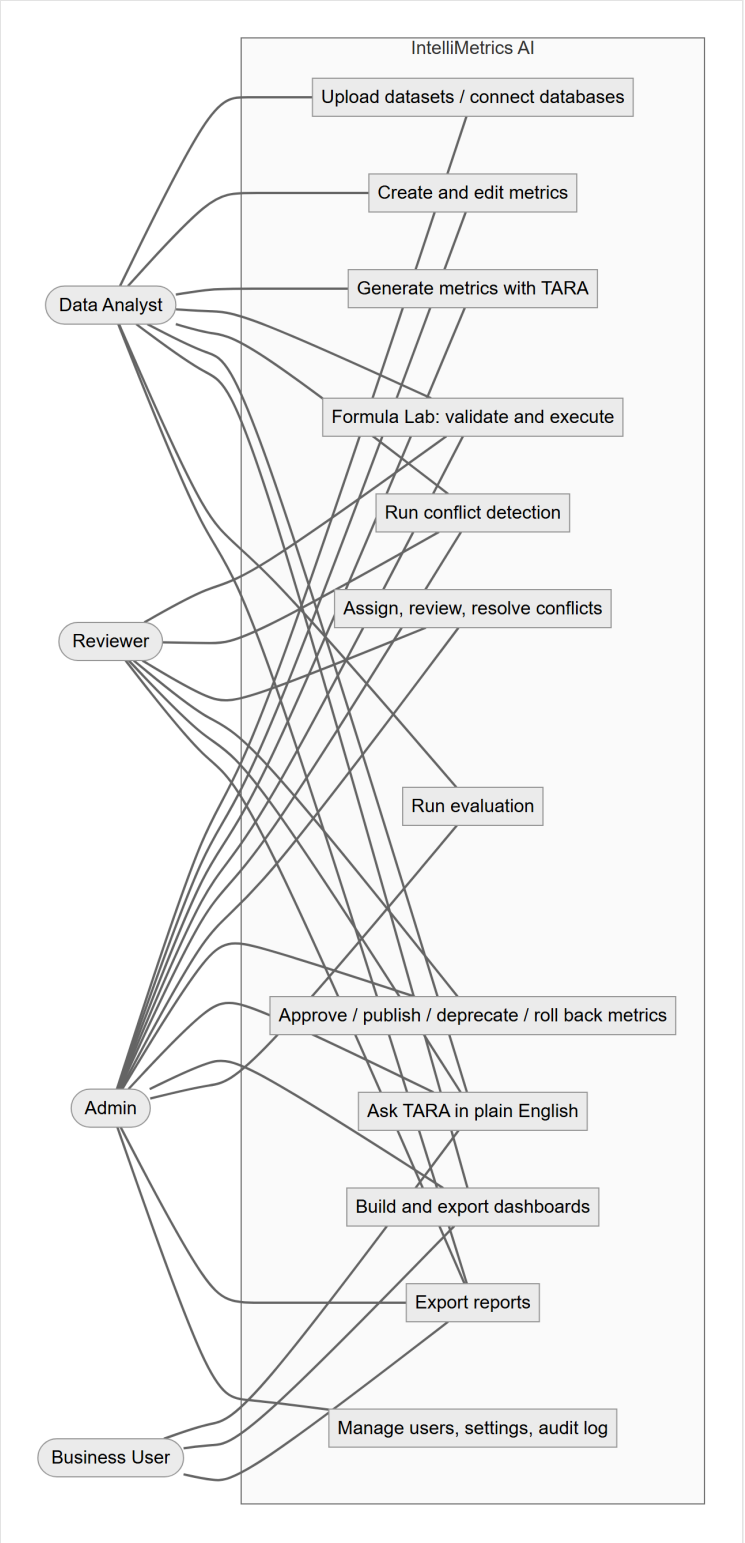
Main dashboard (`overview` router)

Method	Path	Purpose	Capability	Roles allowed
GET	`/api/overview`	overview	-	any signed-in user

Health (`health` router)

Method	Path	Purpose	Capability	Roles allowed
GET	`/health`	health	-	public (no login)

14 Use cases by role



Use cases by role